

MA 573: Advanced Mathematical Foundations of Machine Learning

Course Code	MA 573
Course Name	Advanced Mathematical Foundations of Machine Learning
L-T-P-C	3-1-0-4
Prerequisites	Probability and Statistics, Linear Algebra, and Optimizations
Intended for	B. Tech/ MS/PhD/MSc
Distribution	Elective
Approval	59th BoA (Apr 2025)

Course Contents

- **Mathematical Foundations and Their Learning Systems:** A quick overview of matrix theory, probability distributions, and calculus. Ooceptual role of mathematics in representing data, modeling uncertainty, and enabling learning through optimization. [4 Lectures]
- **Functional Analysis and Kernel Methods:** Functional Analysis in Machine Learning, Matrix norms, submultiplicative properties, Low-rank approximations and Eckart- Young theorem, Introduction to Reproducing Kernel Hilbert Spaces (RKHS): Properties of kernels Mercer's theorem and its implications, Representer theorem, connections to Kernel methods for regression and classification. [8 Lectures]
- **Convex Analysis and Optimization:** Convex Analysis, Gradients, Subgradients and their mathematical properties, Unconstrained Optimization and Convergence of Gradient descent, Constrained Optimization Lagrangian duality and application to machine learning, Theoretical foundations of Regularization, Tikhonov regularization, Sparsity-inducing norms and applications. Connection to Bias-variance tradeoff. [8 Lectures]
- **Kernel Methods and Support Vector Machines:** Kernelized Linear Models: Kernel ridge regression, dual representation, Selection of kernels and hyper-parameters, Support Vector Machines: Analytical formulation, Dual formulation and SMO algorithm, Generalization bounds using VC-dimension and Rademacher complexity. Kernel selection and influence on generalization performance. [6 Lectures]
- **Theoretical Foundations of Neural Networks and Deep Learning Optimization:** Universal approximation theorem, Depth vs. width tradeoffs, Derivation using chain rule, automatic differentiation, Non-convex optimization landscapes: saddle points and local minima, Stochastic gradient descent: convergence analysis. Theoretical properties of Adam, RMSProp. [8 Lectures]
- **Advanced Dimensionality Reduction and Unsupervised Learning:** Expectation Maximization and theoretical foundations Convergence, monotonicity, fixed points, Theory of Clustering: Lloyd's algorithm and convergence, Spectral Clustering and Connection to eigenvalue problems. Evaluation metrics. [8 Lectures]

Laboratory Modules

This course will also include one tutorial session every week dedicated to solving exercises relevant to the studied topics.

Text Books

1. Simovici, D. A., **Mathematical Tools for Data Mining: Set theory, Partial Orders, Combinatorics**, Springer, 2009. (Advanced Information and Knowledge Processing Series)
2. Shai, S. S., and Shai, B. D., **Understanding Machine Learning: Algorithms**, Cambridge University Press, 2014.

References

1. Sra, S., Nowozin, S., and Wright, S. J., **Optimization for Machine Learning**, MIT Press, 2011.
2. Bishop, C. M., **Pattern Recognition and Machine Learning**, Springer, 2006. (Information Science and Statistics Series)
3. Hastie, T., Tibshirani, R., and Friedman, J., **The Elements of Statistical Learning: Data Mining, Inference, and Prediction**, 2nd Edition, Springer, 2009.
4. Boyd, S., and Vandenberghe, L., **Convex Optimization**, Cambridge University Press, 2004.
5. Tan, P. N., Steinbach, M., and Kumar, V., **Introduction to Data Mining**, Pearson, 2014.